

TOI-1516b

R-2077

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI1516_250926 · created 2026-08-09T23:50:37+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460944.776 +/- 0.046 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.162 +/- 0.048
Transit depth (Rp/Rs)^2	2.64 +/- 1.55 [%]
Orbital Inclination (inc)	85.69 +/- 1.35
Airmass coefficient 1 (a1)	3.1 +/- 2.08
Airmass coefficient 2 (a2)	-0.7 +/- 1.69
Scatter in the residuals of the lightcurve fit is	66.63 %
Best Comparison Star	#3 - [392, 242]
Optimal Aperture	3.0
Optimal Annulus	3.5
Transit Duration (day)	0.069 +/- 0.034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.162 ± 0.048 vs 0.1224 (deviation 0.81σ)
Duration fit vs published	0.069 d vs 0.1217 d (deviation 1.53σ)
Mid-time O–C	108.9 min
Depth SNR	3.3
Residual scatter	66.63 %

Light curve

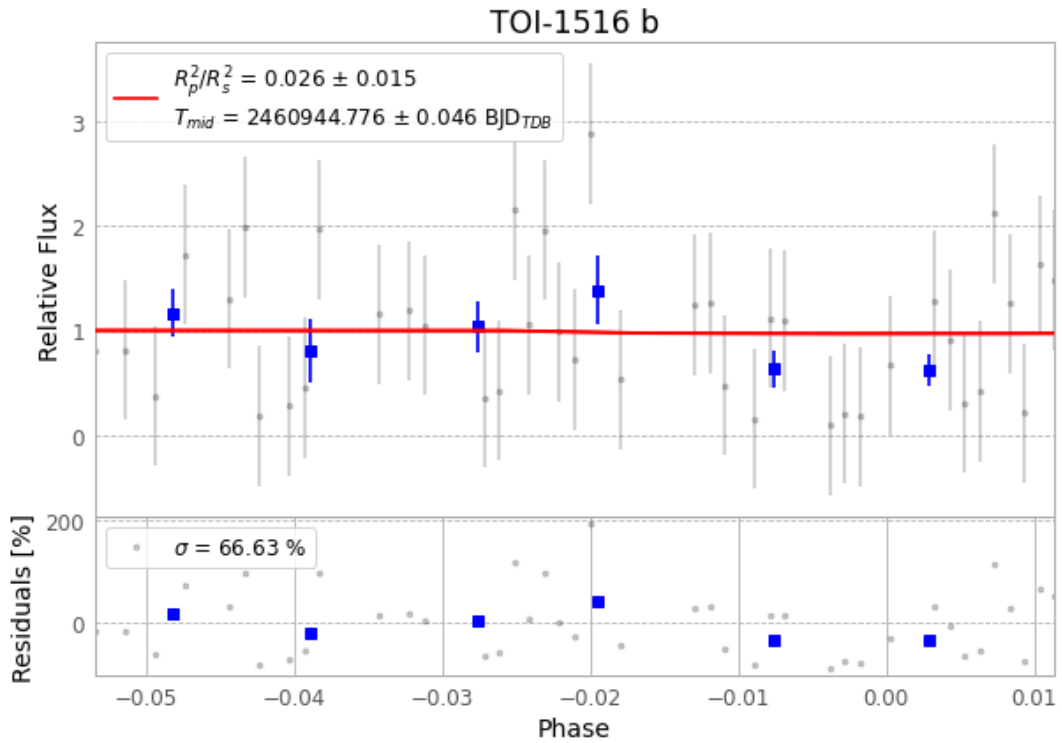


Figure 1 — FinalLightCurve_TOI-1516 b_2025-09-25.png (SHA-256 4aeb6211734fc269...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [591, 423], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3b0c6b9d4bc15277...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

66 FITS frames (43 MB), TOI1516250926035116.FITS → TOI1516250926070915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI1516-250926.log (556b6b344542...), FinalLightCurve_TOI-1516 b_2025-09-25.png (4aeb6211734f...), FinalLightCurve_TOI-1516 b_2025-09-25.csv (f5eaef038b59...)

Compute resources

Wall time 253.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-09 23:50Z.